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② Difference between machine learning and Deep learning

Machine learning

- * A Subset of AI that enables systems to learn from data without explicit programming
- * Performs well with small or structured datasets
- * Uses simpler algorithms such as linear regression, decision trees, and support vector machines (SVM)
- * Generally faster to train due to simpler algorithms and smallest datasets
- * Can be trained on standard CPUs
- * Models are more transparent and easier to interpret

Deep learning

- * A specialized subset of ML that uses multi-layered neural networks to learn
- * Requires larger volumes of unstructured data like images, audio, text
- * Employs complex architectures like convolutional neural network (CNNs) and recurrent neural network (RNNs)
- * Longer training times to ML and images may can train the data
- * Requires high-performance GPUs or TPUs for efficient train
- * Excels in tasks such as image and (NLP) language and automated vehicles.

The Concept of Representation Learning
Representation Learning is a technique in which a neural network automatically learns computational features or on manually designed features.
For example:

- * Lower layers learn edges and textures
- * Middle layers learn shapes and patterns
- * Middle and higher layer, complete objects

Influence of Width

Width refers to the no. of neurons present in a hidden layer

Advantages:

- * Learns more features at each layer
- * Improves the network's ability to model complex relationships

Disadvantages:

Increase memory usage
Requires more computations
may lead to overfitting

3) Compare ReLU, Leaky ReLU (LReLU) and ELU

ReLU

Formula :- $f(x) = \max(0, x)$

Behavior :-
* outputs 0 for negative values
* outputs x for positive value

Advantages :-
* Simple and fast

* Reduces vanishing gradient problem

Disadvantages :-

* suffers from dead neuron problem

Leaky ReLU

$$f(x) = x \quad \text{if } x > 0$$

Advantages :-

* faster and more accurately

Disadvantages :-

It can be slower

ELU

$$f(x) = x \quad \text{if } x > 0$$

Advantages :-

* faster convergence

* Better learning efficiency

Disadvantages :-

* More Image processing systems

* speech recognition of models

Structure

- * visible layer
- * Hidden layer

There are non connecting among neurons within the same layer

Working principle :-

- * Input data is fed into the visible layer
- * Hidden neurons learn latent features
- * The network reconstructs the input data
- * weights are adjusted during contrastive divergence.

Objectives

- * Learn the probability distribution of data
- * Extract meaningful features

Applications

- * Recommendation systems
- * Features extraction
- * Collaborative filtering

Decoder, Encoder Bottleneck (Latent space)
compresses the input data.

Uses in Deep learning

Auto encoder

Structure

An autoencoder contains

- * Encoder
- * Bottleneck
- * Decoder

Working Principle:-

- * Encoder compresses the input data
- * Bottleneck stores important features
- * Decoder reconstructs the original data

Objectives

- * Learn compressed data representation
- * Reduces dimensionality.

Applications

- * Data Compression
- Image denoising
- * Anomaly detection

